

Case study

THE CIO DILEMMA

Observation

The chief observation officer (CIO) of a firm observes that the newly installed Management Information System (MIS) is not being used by middle managers as much as was originally expected. The managers often approach the CIO or some other "computer expert" for help, or worse still, make decisions without facts. "There is surely a problem here," the CIO exclaims.

Information Gathering through Informal Interviews

Talking to some of the middle-level managers, the CIO finds that many of them have very little idea as to what MIS is all about, what kinds of information it could provide, and how to access it and utilize the information.

Obtaining More Information through Literature Survey

The CIO immediately uses the Internet to explore further information on the lack of use of MIS in organizations. The search indicates that many middle-level managers especially the old-timers are not familiar with operating personal computers and experience "computer anxiety." Lack of knowledge about what MIS offers is also found to be another main reason why some managers do not use it.

Formulating a Theory

Based on all this information, the CIO develops a theory incorporating all the relevant factors contributing to the lack of access to the MIS by managers in the organization.

Hypothesizing

From such a theory, the CIO generates various hypotheses for testing, one among them being: Knowledge of the usefulness of MIS would help managers to put it to greater use.

Data Collection

The CIO then develops a short questionnaire on the various factors theorized to influence the use of the MIS by managers, such as the extent of knowledge of what MIS is, what kinds of information MIS provides, how to gain access to the information, and the level of comfort felt by managers in using computers in general, and finally, how often managers have used the MIS in the preceding 3 months.

Data Analysis

The CIO then analyzes the data obtained through the questionnaire to see what factors prevent the managers from using the system.

Deduction

Based on the results, the manager deduces or concludes that managers do not use MIS owing to certain factors. These deductions help the CIO to take necessary action to rectify the situation, which might include, among other things, organizing seminars for training managers on the use of computers, and MIS and usefulness.

Research Language

- Theory
- Conceptualization
- Operationalization
- Variables
- Hypotheses
- Assumptions
- Population
- Sample
- Validity
- Reliability
- Data
- The research process
- Summary
- Methodological queries
- References

Concepts and Constructs

Concept

A *Concept* is a symbol representing an object. It is a property of an object or a certain phenomenon. In other words, it is a generalized idea about an object that has been given a name. Concepts like “democracy”, “commitment”, “participation”, “inflation”, “leadership”, “social mobility”, “justice”, “productivity”, etc. represent some meaning under one general name or label. If we examine these examples more closely, it is evident that each is a word which represents an idea. Generally, these concepts have accepted collection of meaning characteristics associated with certain event, object, condition, situation, or behavior.

Some concepts because of their frequent use have shared meaning and usage. Others, which are less used, do not have shared meaning. Thus, people may hold different concepts of the same thing. Objects, which can be seen, touched, and felt are *concrete concepts*. Those, which cannot be observed and felt, are *abstract concepts*. Their meaning and characteristics are coined depending on the knowledge available in a society. As the knowledge increases, the meaning of these abstract concepts may also change.

Concepts can have many *dimensions*. For instance, “*job satisfaction*” as a concept combines a number of factors which might include pay, promotion, work itself, coworkers, supervisors, and the like. These dimensions can be thought of as different aspects of concepts. After separating out the different dimensions of the concept, the next stage would be to identify which of them are relevant to the research. These dimensions are the *variables* that will be used in research. These variables should capture something about the concept.

Construct

Abstract concepts are called *constructs*. A construct is an image or idea specifically invented for a given research. You can build constructs by combining the similar concepts. Constructs are thus complex and imply condition of an organism whose existence can only be deduced from observable phenomenon, rather than by direct observation (Cooper & Schindler, 2011). Both Concepts and constructs are abstract. However, constructs are higher-level abstractions. Abstract concepts are thus those constructs that are completely independent of a specific time and Place. They are often developed by and closely associated with particular types of research. Thus, the constructs need to be translated into observable incidents or operation to produce empirical evidence